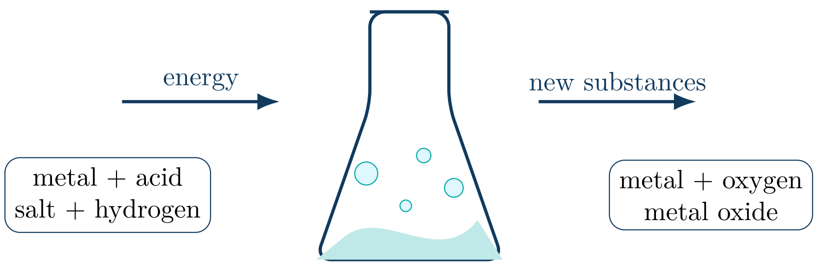
Chapter 8

Chemical Reactions

Student Active Fill-In Notes

IGCSE Year 8 Science Foundation



What this booklet helps students do: fill in key scientific words during class, explain energy changes, write word equations, compare metal reactivity, predict products, and describe key observations using accurate scientific language.

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Chapter Overview

Learning Goals

By the end of this chapter, students should be able to:

- distinguish between exothermic and endothermic reactions;
- describe how metals react with oxygen, water, and dilute acids;
- write and complete simple word equations;
- use observations such as fizzing, colour change, and temperature change as evidence;
- use the reactivity series to predict whether a reaction will happen;
- explain displacement reactions using the idea of a more reactive metal replacing a less reactive metal.

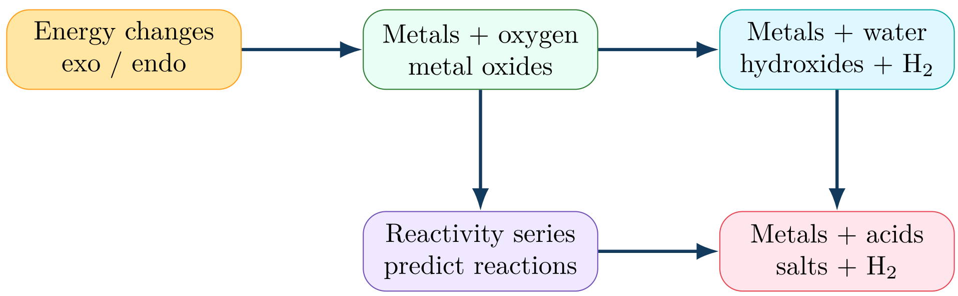
How to Use These Notes

Student first	Fill in the missing keywords during the lesson. Do not copy passively; listen, predict, then write.
Tutor first	Pause at each blank and ask the student to predict the word before revealing it. Use the purple teaching notes as the lesson flow.
Before exams	Cover the answer-key version and re-fill the student version from memory. Re-practise word equations until product names are automatic.

Student Action

When you see a blank, first try to say the answer aloud. Then write it neatly on the line. The blanks focus on exam keywords, product names, gases, observations, and reaction rules.

Diagram: The Five Key Ideas in Chapter 8



1 8.1 Exothermic and Endothermic Reactions

8.1 Energy in chemical reactions

Big Idea: Energy transfer

Every chemical reaction involves an ________________. Some reactions ________________ to the surroundings, while other reactions _________________ from the surroundings.

Student Study Notes

______________________ – a reaction that ___________ heat energy to the surroundings. The surroundings become __________.

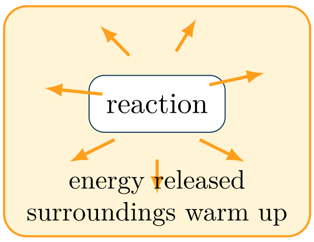
_______________________ – a reaction that ___________ energy from the surroundings. The surroundings become __________.

- In an ______________ reaction, energy moves ______ of the reacting chemicals and into the surroundings.
- In an ________________ reaction, energy moves _______ the reacting chemicals from the surroundings.
- A temperature ___________ usually suggests an ______________ reaction.
- A temperature ___________ usually suggests an ________________ reaction.

Type	What happens to surroundings?	Examples
Exothermic	Temperature ____________	burning fuels, neutralisation, many metal-acid reactions
Endothermic	Temperature ____________	thermal decomposition, photosynthesis, some dissolving reactions

Diagram: Energy flow in exothermic and endothermic reactions

Exothermic



Endothermic

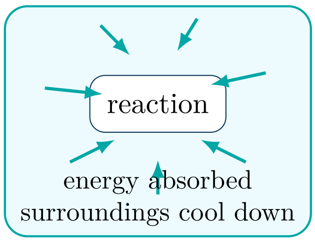
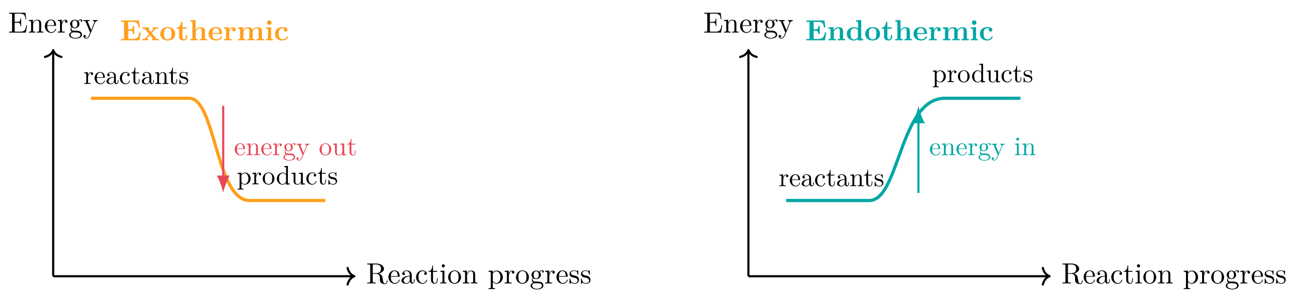


Diagram: Simple energy level sketches



Tutor Teaching Notes

1. Start with a real feeling: let the student carefully feel the outside of a container before and after a safe demonstration.
2. Demonstrate an endothermic example such as baking soda with vinegar. Ask: "What happened to the temperature?"
3. Demonstrate an exothermic example such as magnesium with dilute hydrochloric acid, using a very small amount and proper safety equipment.
4. Link the words to memory: **exo** sounds like **exit**; energy exits the reaction. **endo** sounds like **enter**; energy enters the reaction.
5. Ask the student to explain the difference using the words *surroundings*, *temperature*, *released*, and *absorbed*.

Quick Check

1. A reaction makes the test tube feel warm. Is it exothermic or endothermic?
2. Explain why an instant cold pack is an example of useful energy transfer.
3. Complete the sentence: In an endothermic reaction, energy is ______________ from the surroundings.
4. Why is measuring temperature better than only touching the beaker?

Common Mistakes to Avoid

Do not say "the reaction becomes cold" without explaining the surroundings. A stronger answer says: **the reaction absorbs energy from the surroundings, so the surroundings become colder.**

2 8.2 Reactions of Metals with Oxygen

8.2 Oxidation and metal oxides

Big Idea: Metals can gain oxygen

When many metals are heated in air, they react with __________ to form ______________. This is called ____________ because the metal ______________.

Word Equation: Metals with oxygen

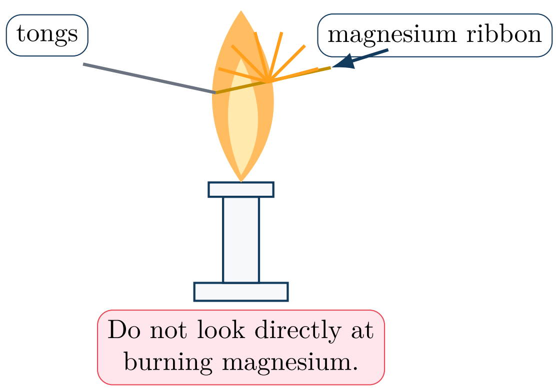
________ + __________ → ______________

Student Study Notes

- A ______________ is a compound made from a ________ and __________.
- Different metals react with oxygen at different ________ and ____________.
- ____________ burns with a very __________________ and forms white __________________ powder.
- _______ usually glows _________ and forms an oxide layer more ________.
- __________________ are very ____________, so they do not easily react with oxygen even when heated.

Metal	Observation in oxygen/air	Product
Magnesium	________________; white powder forms	________________
Iron	glows ________; oxide layer forms slowly	____________
Copper	____________ forms when heated	______________
Gold	no visible reaction	no oxide under normal classroom conditions

Diagram: Heating a metal in air



Safety First

Burning magnesium gives off an extremely bright white light. Students must wear safety goggles and must not stare directly at the flame. Use a heat-proof mat and keep flammable materials away.

Tutor Teaching Notes

1. Begin with a prediction: "Will all metals burn like magnesium?"
2. Demonstrate heating magnesium ribbon. Ask students to record observations **before**, **during**, and **after** the reaction.
3. Write the equation slowly: **Magnesium + Oxygen → Magnesium Oxide**.
4. Compare with rusty iron or tarnished copper to show that oxidation can be fast or slow.
5. Challenge question: If zinc reacts with oxygen, what is the product called?

Quick Check

1. What is the general word equation for a metal reacting with oxygen?
2. Why is magnesium oxide called an oxide?
3. A metal forms a black coating when heated in air. What evidence suggests a chemical reaction has occurred?
4. Predict the product: Calcium + Oxygen → ____________________

Exam Tip

In word equations, the product name usually combines the metal name with the non-metal part. For oxygen reactions, the product ends with **oxide**.

3 8.3 Reactions of Metals with Water

8.3 **Hydrogen gas and metal hydroxides**

Big Idea: Reactive metals can react with water

Some metals react with cold water to produce a ________________ and ______________. The more reactive the metal, the ________ and more _________ the reaction.

Word Equation: Metals with water

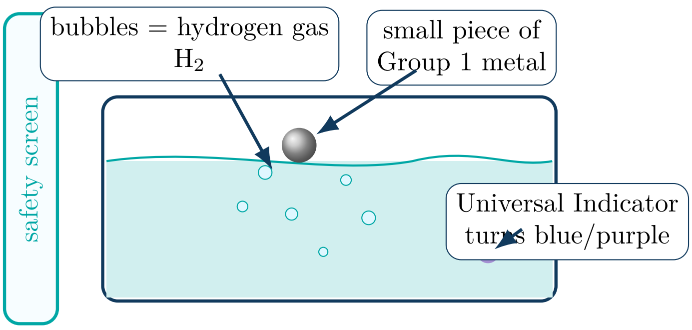
_________ + _________ → _____________________ + _____________

Student Study Notes

- __________ metals are very reactive with water.
- The bubbles produced are ______________.
- The solution becomes __________ because a _________________ forms.
- Universal Indicator can show this change by turning _______________.
- Lithium reacts ____________, sodium reacts more ___________, and potassium reacts ____________________.
- Magnesium does not react much with ____________, but it reacts more easily with ________.

Metal	Observation with cold water	Relative reactivity
Lithium	floats and ______________	reactive
Sodium	melts into a _____________ and moves quickly on the surface	more reactive
Potassium	reacts _____________; may ignite hydrogen with a _____________	most reactive of these three
Magnesium	almost no visible reaction with ____________	much less reactive

Diagram: Group 1 metal reacting with water



Safety First

Group 1 metals must be handled by the teacher or tutor only, using very small pieces and a safety screen. Never touch these metals with wet hands.

Tutor Teaching Notes

1. Put Universal Indicator in the water before the demonstration. This gives visual evidence that a metal hydroxide forms.
2. Ask students to observe the speed of fizzing, movement, melting, flame colour, and temperature change.
3. Teach the hydrogen test: a lighted splint gives a **squeaky pop** if hydrogen gas is present.
4. Let students rank lithium, sodium, and potassium based only on the observations.
5. Drill incomplete equations such as: Sodium + Water → ______________ + ____________.

Quick Check

1. What two products form when a reactive metal reacts with water?
2. Why does Universal Indicator turn blue or purple?
3. Which is more reactive with water: lithium or potassium? Give one observation.
4. Complete: Potassium + Water → ________________ + ____________

Common Mistakes to Avoid

Do not write *metal oxide* as the product when a metal reacts with water. With water, the product is usually a **metal hydroxide** and **hydrogen**.

4 8.4 The Reactivity Series of Metals

8.4 Predicting reactions

Big Idea: Reactivity is a ranking system

The ____________________ places metals in order from ________________ to ________________. It helps us __________ how metals behave with oxygen, water, acids, and metal salt solutions.

Student Study Notes

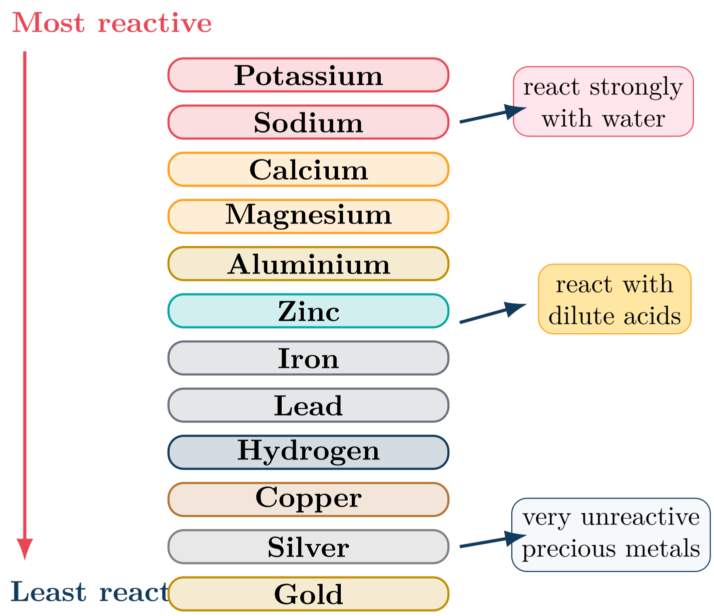
- Metals near the ______ are very reactive. They react strongly with water and oxygen.
- Metals in the _________ usually react with acids, but not with cold water.
- Metals near the __________ are unreactive. Copper, silver, and gold do not react with dilute acids.
- In a _______________________, a ________________ metal can replace a ________________ metal from its compound.

Word Equation: Displacement example

_______________ + ____________________ → _________________________ + ____________

The magnesium is ________________ than copper, so magnesium takes copper's place in the compound. ___________ metal is produced.

Diagram: A Year 8 reactivity series ladder



Memory Mnemonic

Please Stop Calling Me A Careless Zebra, Instead Try Learning How Copper Saves Gold.

This can stand for: Potassium, Sodium, Calcium, Magnesium, Aluminium, Carbon, Zinc, Iron, Tin, Lead, Hydrogen, Copper, Silver, Gold. For Year 8, focus especially on metals and hydrogen.

Diagram: Displacement rule

More reactive metal **can displace** $\xrightarrow{\textbf{pushes out}}$ Less reactive metal **cannot displace**

Example: magnesium is above copper,
so magnesium displaces copper from copper sulfate.

Tutor Teaching Notes

1. Do not begin with memorisation only. First, show evidence: metals behave differently with water and acids.
2. Let the student build a mini series from observations, then reveal the standard order.
3. Use coloured cards for metals. Ask: "Which metal is higher? Which one can replace the other?"
4. For displacement, set up a table with solid metals against salt solutions. The student ticks reactions and crosses no reaction.
5. Emphasise the decision rule: **solid metal higher in the series = reaction occurs**.

Quick Check

1. Which is more reactive: zinc or copper?
2. Will copper displace magnesium from magnesium sulfate? Explain.
3. What is a displacement reaction?
4. Complete: Zinc + Copper Sulfate → __________________ + _____________

Common Mistakes to Avoid

A common error is thinking that any metal will react with any salt solution. The correct rule is: a metal only displaces another metal if it is **more reactive** than the metal in the compound.

5 8.5 Reactions of Metals with Dilute Acids

8.5 Salts and hydrogen gas

Big Idea: Acids produce salts with reactive metals

When a reactive metal reacts with a dilute acid, the products are a ________ and ________________. The name of the salt depends on the ________ used.

Word Equation: Metals with dilute acids

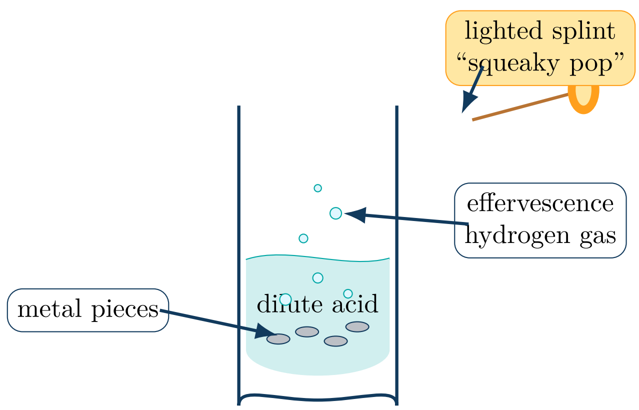
__________ + _________ → _________ + _______________

Student Study Notes

- Fizzing or bubbling is called ________________. It is caused by ________________ escaping.
- Many metal-acid reactions are ______________, so the test tube may feel ________.
- Metals ________________ in the reactivity series react with dilute acids.
- Metals ________________, such as copper, silver, and gold, do not react with dilute acids.
- The salt name depends on the acid:

Acid used	Salt ending	Example
Hydrochloric acid	___________	zinc chloride
Sulfuric acid	___________	magnesium sulfate
Nitric acid	___________	calcium nitrate

Diagram: Metal + acid reaction and hydrogen test



Tutor Teaching Notes

1. Let students compare magnesium, zinc, iron, and copper with dilute hydrochloric acid.
2. Ask them to rank reactivity by speed of bubbling: fastest fizzing means more reactive.
3. Teach the hydrogen test using a small gas sample only. Hydrogen burns with a squeaky pop.
4. Drill salt names until automatic: hydrochloric acid gives chlorides, sulfuric acid gives sulfates, nitric acid gives nitrates.

5. Ask students to explain why copper does not react with dilute acid using the reactivity series.

Safety First

Use dilute acids only. Wear eye protection. Keep the lighted splint away from the main reaction tube and test only small amounts of collected hydrogen gas.

Quick Check

1. What gas is produced when zinc reacts with dilute hydrochloric acid?
2. What is the test for hydrogen gas?
3. Complete: Magnesium + Sulfuric Acid → ____________________ + _____________
4. Why does copper not react with dilute hydrochloric acid?

Exam Tip

When naming salts, first identify the acid. Then add the correct ending: hydrochloric acid → chloride, sulfuric acid → sulfate, nitric acid → nitrate.

6 Practical Skills: What Students Should Observe

Observation Language Bank

Use precise words instead of vague words like "something happened".

Observation	Strong scientific wording
Bubbles	__________________ was observed, showing a _______ was produced.
Warm test tube	The temperature of the surroundings _____________, suggesting an _______________ reaction.
Cold beaker	The temperature of the surroundings _____________, suggesting an ________________ reaction.
Colour change	A new substance may have formed.
Solid forms	A precipitate or metal deposit may have formed.
Bright flame	Energy was ____________ as light and heat.

Diagram: From observation to conclusion

Observation What did you see? → Evidence What does it show? → Conclusion What reaction occurred?

Mini Practical Template

Question	Which metal reacts fastest with dilute hydrochloric acid?
Variables	Keep the acid concentration, acid volume, metal size, and temperature the same.
Method	Add equal-sized pieces of different metals to equal volumes of acid. Record the speed of bubbling.
Result	Faster bubbling means faster hydrogen production.
Conclusion	The metal with the fastest bubbling is the most reactive of the metals tested.

7 Chapter Summary Tables

Core Word Equations

Reaction type	General word equation
Metal with oxygen	________ + __________ → ______________
Metal with water	________ + _________ → __________________ + ____________
Metal with dilute acid	________ + ________ → _______ + ____________
Displacement reaction	____________________ + __________________________ → New Compound + Less Reactive Metal

Reaction Clues

Clue	Possible meaning
Temperature rises	____________ reaction
Temperature falls	_____________ reaction
Bubbles form	______ is produced
Squeaky pop	______________ is present
Universal Indicator turns blue/purple	________ forms, such as a metal hydroxide
A new solid appears	A new substance forms; possible displacement or precipitation

Salt Naming Helper

Acid	Salt ending	Example product
Hydrochloric acid	__________	magnesium chloride
Sulfuric acid	__________	zinc sulfate
Nitric acid	__________	calcium nitrate

8 Active Keyword Tracker

Fill These from Memory

After teaching, ask the student to close the notes and complete this page from memory.

1. A reaction that releases heat: ________________
2. A reaction that absorbs heat: ________________
3. Metal + oxygen gives: __________________
4. Metal + water gives: ______________________________
5. Metal + acid gives: ______________________
6. Bubbles are called: ___________________
7. Hydrogen test: __________________
8. Ranking of metals: _____________________
9. More reactive metal replaces a less reactive metal: ____________________
10. Hydrochloric acid makes a: ______________
11. Sulfuric acid makes a: ______________
12. Nitric acid makes a: ______________

9 Student Practice Worksheet

Part A: Key Words

Match each keyword to the correct meaning.

1. Exothermic
2. Endothermic
3. Oxidation
4. Displacement
5. Effervescence

A	A more reactive metal replaces a less reactive metal in a compound.
B	A reaction where energy is absorbed from the surroundings.
C	Bubbles forming because a gas is produced.
D	A reaction where energy is released to the surroundings.
E	A reaction where a substance gains oxygen.

Part B: Complete the Word Equations

1. Magnesium + Oxygen → ______________________________
2. Sodium + Water → _____________________ + _______________
3. Zinc + Hydrochloric Acid → _____________________ + _______________
4. Magnesium + Copper Sulfate → _____________________ + _______________
5. Calcium + Nitric Acid → _____________________ + _______________

Part C: Explain Your Thinking

Answer in full sentences.

1. A student adds copper to dilute hydrochloric acid. There is no fizzing. Explain why.
2. Potassium reacts more violently with water than lithium. What does this tell us about potassium's reactivity?
3. A reaction vessel becomes colder during a reaction. Explain the energy transfer.
4. Why is the squeaky pop test useful in metal-acid reactions?

10 One-Page Revision Checklist

Before the Lesson Ends, Can the Student Do These?

- Define exothermic and endothermic.
- Use temperature change as evidence.
- Write: Metal + Oxygen → Metal Oxide.
- Name metal oxides correctly.
- Write: Metal + Water → Metal Hydroxide + Hydrogen.
- Explain why Universal Indicator changes colour.
- Use the squeaky pop test for hydrogen.
- Recall the key order of the reactivity series.
- Predict simple displacement reactions.
- Write: Metal + Acid → Salt + Hydrogen.
- Name chloride, sulfate, and nitrate salts.
- Explain why copper does not react with dilute acids.

Chemistry becomes easier when every reaction is linked to three things:

observation → word equation → reactivity explanation.